WORKSHEET

Chapter 4: Applications of recurrence relations

1 Michael's computer cost $1750 and depreciates at a rate of 23% per year.

a Write a recurrence relation to represent the value of Michael's computer.

b In how many years will the value of the computer be less than $500?

2 Jana enters a 100-km charity walk. She walks at a speed of 6.4 km/h.

a Write a recurrence relation to represent how far Jana has left to walk.

b How many hours will it take Jana to complete the walk?

3 A cherry farmer plants new trees every year. He had 278 trees in 2011, 303 trees in 2012 and 328 trees in 2013.

a Write a recurrence relation to represent the number of cherry trees on the farm.

b If he continues to plant new trees at this rate, how many trees should the farmer have in 2017?

4 Cassie invests $80 000 for 8 years. Each year, the investment grows by 7.5%.

a Write a recurrence relation to represent Cassie's investment.

b Calculate the value of Cassie's investment at the end of the 8 years, to the nearest dollar.

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5 As a result of a rabbit plague, a species of fern in a small forest is expected to become endangered. The number of ferns in the forest per year now follows the recurrence relation:

$$t_{n+1} = 0.8t_n + 125$$

This year there are 13 980 ferns.

a List the number of ferns expected to be in the forest in each of the next 5 years. Round each answer to the nearest whole number.

b In the long term, how many ferns are expected to be in the forest?

6 The recurrence relation:

$$t_{n+1} = 0.64t_n + 76$$

describes the population of frogs in an area of bush.

a If the initial population is 320, what will the population be 3 years later? Answer to the nearest whole number.

b What is expected to be the long-term population of frogs in this area? Answer to the nearest whole number.

7 Omid invests $10 000. The investment grows by 10% annually, and he withdraws $1500 once a year to buy gifts for his family.

a Write a recurrence relation to represent the value of Omid's investment.

b What is the value of Omid's investment after 5 years?

c How much money has Omid withdrawn from the account over this time?

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d If Omid was not receiving interest on this money, how much would he have left of the original $10 000?

8 A heart attack victim is given a blood-thinning drug. After an initial dose of 55 milligrams, the concentration of the drug in the bloodstream reduces by 45% each day and must be topped up daily with another dose of 15 mg.

a Write a recurrence relation for the quantity of the drug in the bloodstream.

b List the amount (in mg) of the drug in the bloodstream over the first 5 days. Round answers to two decimal places.

c What is the steady-state value of the drug concentration?

9 Marica borrows $26 000 from a credit union to buy a car. Each month, she is charged 1% interest on the amount owing, and Marica repays $450.

a Write a recurrence relation for the amount of money Marica owes the bank each month.

b By how much is her loan reduced after her first repayment? (Hint: Her first repayment is made immediately after the interest owing is added to the loan.)

c How much does Marica owe the credit union after 6 months? Answer to the nearest dollar.

10 The equation:

$$t_{n+1} = rt_n + d, t_0 = a$$

describes an investment made by Halal, who invests $2600 every month and earns 1.5% interest monthly on his investment. His initial deposit into the account was $5000.

a Find the values of r, d and a.

b Find the value of the investment after each of the first 6 months. Answer to the nearest cent.

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c How long does it take for the value of the investment to exceed $20 000?

11 The water added to a tropical fish tank is initially 18°C. After 1 hour, it is 18.8°C, and after 2 hours it is 19.52°C. The recurrence relation representing the temperature of the water in the tank over time is:

$$t_{n+1} = rt_n + d, t_0 = a.$$

a Find the values of r, d and a.

b Find the temperature of the water in the tank after 5 hours, to one decimal place.

c What will be the long-term temperature of the water?

12 A population of 32 mice grows to 76 mice in 1 month and to 186 mice the following month. The mouse population can be represented by the recurrence relation:

$$t_{n+1} = rt_n + d, t_0 = a.$$

a Find the values of r, d and a.

b Use the recurrence relation to find the size of the population after 6 months, to the nearest whole number.

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Answers

1 **a** $t_{n+1} = 0.77t_n$, $t_0 = 1750$

b 5 years

2 **a** $t_{n+1} = t_n - 6.4$, $t_0 = 100$

b 16 hours

3 **a** $t_{n+1} = t_n + 25$, $t_1 = 278$

b 428

4 **a** $t_{n+1} = 1.075t_n$, $t_0 = 80\,000$

b \$142 678

5 **a** 11 309, 9172, 7463, 6095, 5001

b 625

6 **a** 240

b 211

7 **a** $t_{n+1} = 1.1t_n - 1500$, $t_0 = 10\,000$

b \$6947.45

c \$7500

d \$2500

8 **a** $t_{n+1} = 0.55t_n + 15$, $t_1 = 55$

b 55, 45.25, 39.89, 36.94, 35.32

c 33.3 mg

9 **a** $t_{n+1} = 1.01t_n - 450$, $t_0 = 26\,000$

b \$190

c \$24 831

10 **a** $r = 1.015$, $d = 2600$, $a = 5000$

b \$7675, \$10 390.13, \$13 145.98, \$15 943.17, \$18 782.31, \$21 664.05

c 6 months

11 **a** $r = 0.9$, $d = 2.6$, $a = 18$

b 21.3°C

c 26°C

12 **a** $r = 2.5$, $d = -4$, $a = 32$

b 7164

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